MIL-O-44202A
14 October 1988
SUPERSEDING
MIL-O-44202
15 January 1986

MILITARY SPECIFICATION

OMELET WITH HAM, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized omelet with ham in flexible pouches for use as a component of the Meal, Ready-To-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Grades of Butter

(Application for copies should be addressed to the Dairy Standardization Section, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2750, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

* AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

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3.2 Ingredients. All ingredients shall be clean, sound, wholesome and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Ham, canned or cooked. The ham shall be high commercial grade. The ham shall not be labeled "water added", and shall not contain dextrose. The canned or cooked ham shall be maintained at an internal temperature of 28° to 40°F for a period of time not to exceed 60 days from the date of pack until preparation of the finished product.

* 3.2.2 Whole eggs, liquid or frozen. Whole eggs may be liquid or frozen and shall be processed and labeled in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The whole eggs shall be egg whites and egg yolks in their natural proportions as broken directly from the shell eggs, as evidenced by a USDA Egg Products Inspection Certificate. For liquid whole eggs, the certificate shall state the date and time of pasteurization. Liquid whole eggs shall be held at an internal temperature of 28° to 40°F for not more than 72 hours from the time of pasteurization until the start of formulation of the product in which they are used. Frozen whole eggs shall be held at 10°F or lower and used within 120 days from the date of certification. Frozen whole eggs shall be thawed and held at an internal temperature of 28° to 40°F and shall be used in the formula within 24 hours after they are thawed. The whole eggs shall be free from off-odors, off-flavors, or foreign odors and flavors, such as metallic, musty, sour, fruity, or sulfide-like, and shall be used in the formula within 24 hours after they are thawed.

3.2.3 Water. Water used for formulation, ice making and washing shall conform to the National Primary Drinking Water Regulations.

3.2.4 Butter, lightly salted. Butter shall be lightly salted and shall be U.S. Grade A or better as specified in the U.S. Standards for Grades of Butter.

* 3.2.5 Hominy grits. Hominy grits shall be made from white corn from which the bran, germ, and caps have been removed. The hominy grits shall have a relatively bland flavor, a mild cereal aroma, a moisture content of not more than 13.5 percent, and a fat content of not more than 2.0 percent. The hominy grits shall be of a size commonly used for breakfast cereal.

3.2.6 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.7 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

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3.2.8 Citric acid. The citric acid shall comply with the Food Chemicals Codex.

3.2.9 Color, annatto, dry. The annatto color shall be a reddish-orange water soluble powder of the extractive of Annatto bixa orellana. The powder shall contain less than 5 percent moisture. One hundred percent of the powder shall pass through a U.S. No. 10 sieve. The total color (as percent Norbixin) shall be 15 percent.

3.3 Preparation and processing. Processing shall be on a continuous basis.

* 3.3.1 Preparation of ham. The ham shall be ground once through a plate having holes 0.5 inch in diameter. The ground ham shall be held for not more than 24 hours at an internal temperature of 28° to 40°F prior to preparation of the omelet with ham.

3.3.2 Preparation of omelet with ham. The omelet with ham shall be formulated and prepared as follows:

Ingredient	Percent by weight
Eggs, whole	45.000
Ham, ground	36.800
Water	11.876
Butter	3.000
Hominy grits	3.000
Lecithin	0.200
Pepper, white, ground	0.070
Citric acid	0.050
Annatto powder	0.004

NOTE: All mixing shall be done in a manner that will minimize the incorporation of air into the product.

* a. The annatto powder shall be completely dissolved in a small amount of the formula water.

b. The water, hominy grits, butter, and lecithin shall be cooked in a steam jacketed kettle until the mixture has fully thickened to form a cereal.

c. The volume of the cereal shall be adjusted with water to compensate for evaporation loss.

d. The ground ham shall be uniformly mixed with the cooked cereal.

e. To assure an even distribution of the annatto coloring, the remaining ingredients shall be added to the ham/cereal mixture in the following order, eggs, dissolved annatto, white pepper, and citric acid. The ingredients shall be mixed only to the extent necessary to assure a uniform blend of all the ingredients.

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f. If not immediately filled, the mixed product shall be maintained in the temperature range of 28° to 40°F and filled into pouches within 4 hours after preparation.

3.4 Pouch filling and sealing. Each pouch shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. The prepared omelet with ham shall be filled into pouches using a single stage filling operation.

b. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements for class 3 as specified in MIL-P-44073.

c. A hot fill system is not permitted.

d. Each filled and sealed pouch shall be in the retort process within 2 hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 3 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to, glass, dirt, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, moldy, rancid, sour, or stale.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 6.0 ounces.

e. No individual pouch shall contain less than 5.5 ounces of product.

f. No individual pouch shall contain more than 0.6 ounces by weight of free liquid.

g. Ham particles shall be dispersed evenly throughout the omelet.

h. Texture of the omelet shall not be rubbery.

i. The average fat content shall not be greater than 12.0 percent.

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j. No individual pouch of product shall have a fat content greater than 14.0 percent.

k. No individual pouch of product shall have a salt content greater than 1.3 percent or less than 0.5 percent.

l. The product shall show no evidence of excessive heating (materially darkened or scorched).

* m. Annatto color shall be uniform throughout the omelet.

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The egg products component, ham components and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

* 4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved products.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 5.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 6.0 ounces.

* 4.5.5 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 180° to 190°F water for 10 minutes, opened, and inspected for the defects indicated in table I. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

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TABLE I. Product defects 1/ 2/

Category	Defect
Major	
101	Free liquid in a pouch weighs more than 0.6 ounces. 3/
102	Ham particles not evenly dispersed throughout the omelet. 4/
103	Texture of product is rubbery. 4/
104	Product shows evidence of excess heating (materially darkened or scorched). 4/
* 105	Annatto color not uniform throughout the omelet.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall only be performed when deemed necessary by an Agricultural Marketing Service (AMS) agent).

3/ To determine amount of free liquid in the pouch, the sealed pouch shall be heated in 180° to 190°F water for 10 minutes. Remove the pouch from the hot water, cut top of pouch and drain any free liquid into a suitable, tared container. Report weight of free liquid to the nearest 0.1 ounce.

4/ Examination for appearance and texture conformance of the omelet shall be made following the determination of free liquid. The omelet shall be cooled and then split horizontally to the flat surface to yield two approximately equal halves. Both sides of the omelet shall be visually examined for defect 102.

4.5.6 Fat and salt content testing. The unopened sample pouches shall be warmed in a water bath to melt fat adhering to the inside of the pouches. The sample pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring blender or equivalent. The fat and salt content of the product from each pouch in the sample shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt

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requirement in 3.6 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with all the requirements of MIL-P-44073.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

* 6.3 Subject term (key word) listing.

Ham
Meal, Ready-To-Eat
Omelet
Ration
Thermostabilized

6.4 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0648

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-44202A	Omelet w|Ham, Thermostabilized, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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